

Counting

Chapter 6

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Chapter Summary

- The Basics of Counting
- The Pigeonhole Principle (鴿籠原理)
- Permutations(排列) and Combinations(組合)
- Binomial Coefficients(二項式係數)and Identities (等式)
- Generalized Permutations and Combinations
- Generating Permutations and Combinations

The Basics of Counting

Section 6.1

Basic Counting Principles: The Product Rule

The Product Rule: A procedure can be broken down into a sequence of two tasks. There are n_1 ways to do the first task and n_2 ways to do the second task. Then there are $n_1 \cdot n_2$ ways to do the procedure.

Example: How many bit strings of length seven are there?

Solution: Since each of the seven bits is either a 0 or a 1, the answer is $2^7 = 128$.

The Product Rule

Example: How many different license plates can be made if each plate contains a sequence of 3 uppercase English letters followed by 3 digits?

Solution: By the product rule,

$$\text{there are } \underbrace{26 \cdot 26 \cdot 26}_{\substack{26 \text{ choices} \\ \text{for each} \\ \text{letter}}} \cdot \underbrace{10 \cdot 10 \cdot 10}_{\substack{10 \text{ choices} \\ \text{for each} \\ \text{digit}}} = 17,576,000$$

different possible license plates.

Counting Functions

Counting Functions: How many functions are there **from** a set with m elements **to** a set with n elements?

Solution: Since a function represents a choice of one of the n elements of the codomain for each of the m elements in the domain, the product rule tells us that there are $n \cdot n \cdots n = n^m$ such functions.

Counting One-to-One Functions: How many one-to-one functions are there **from** a set with m elements **to** one with n elements?

Solution: Suppose the elements in the domain are $a_1, a_2, \dots, a_m$. There are n ways to choose the value of a_1 and $n-1$ ways to choose a_2 , etc. The product rule tells us that there are $n(n-1)(n-2)\cdots(n-m+1)$ such functions.

Telephone Numbering Plan

(Reading Exercise)

Example: The *North American numbering plan (NANP)* specifies that a telephone number consists of 10 digits, consisting of a 3-digit area code, a 3-digit office code, and a 4-digit station code. There are some restrictions on the digits.

- Let X denote a digit from 0 through 9.
- Let N denote a digit from 2 through 9.
- Let Y denote a digit that is 0 or 1.
- In the old plan (in use in the 1960s) the format was $NYX-NNX-XXX$.
- In the new plan, the format is $NXX-NXX-XXX$.

How many different telephone numbers are possible under the old plan and the new plan?

Solution: Use the Product Rule.

- There are $8 \cdot 2 \cdot 10 = 160$ area codes with the format NYX .
- There are $8 \cdot 10 \cdot 10 = 800$ area codes with the format NXX .
- There are $8 \cdot 8 \cdot 10 = 640$ office codes with the format NNX .
- There are $10 \cdot 10 \cdot 10 \cdot 10 = 10,000$ station codes with the format $XXXX$.

Number of old plan telephone numbers: $160 \cdot 640 \cdot 10,000 = 1,024,000,000$.

Number of new plan telephone numbers: $800 \cdot 800 \cdot 10,000 = 6,400,000,000$.

Counting Subsets of a Finite Set

Counting Subsets of a Finite Set: Use the product rule to show that the number of different subsets of a finite set S is $2^{|S|}$. (*In Section 5.1 Ex10, mathematical induction was used to prove this same result.*)

Solution: When the elements of S are listed in an arbitrary order, there is a **one-to-one correspondence** between **subsets of S** and **bit strings of length $|S|$** . When the i th element is in the subset, the bit string has a 1 in the i th position and a 0 otherwise.

By the product rule, there are $2^{|S|}$ such bit strings, and therefore $2^{|S|}$ subsets.

Product Rule in Terms of Sets

- If $A_1, A_2, \dots, A_m$ are finite sets, then the number of elements in the **Cartesian product** (see p.123) of these sets is the product of the number of elements of each set.
- The task of choosing an element in the Cartesian product $A_1 \times A_2 \times \dots \times A_m$ is done by choosing an element in A_1 , an element in A_2 , ..., and an element in A_m .
- By the product rule, it follows that:
$$|A_1 \times A_2 \times \dots \times A_m| = |A_1| \cdot |A_2| \cdot \dots \cdot |A_m|.$$

Basic Counting Principles: The Sum Rule

The Sum Rule: If a task can be done either in one of n_1 ways or in one of n_2 ways, where none of the set of n_1 ways is the same as any of the n_2 ways, then there are $n_1 + n_2$ ways to do the task.

Example: The mathematics department must choose either a student or a faculty member as a representative for a university committee. How many choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student.

Solution: By the sum rule it follows that there are $37 + 83 = 120$ possible ways to pick a representative.

The Sum Rule in terms of sets.

- The sum rule can be expressed in terms of sets.

$$|A \cup B| = |A| + |B|$$

as long as A and B are disjoint sets.

- Or more generally,

$$|A_1 \cup A_2 \cup \cdots \cup A_m| = |A_1| + |A_2| + \cdots + |A_m|$$

when $A_i \cap A_j = \emptyset$ for all i, j .

Combining the Sum and Product Rule

Example: Suppose statement **labels** in a programming language can be **either a single letter or a letter followed by a digit**. Find the number of possible labels.

Solution: Use the product rule.

$$26 + 26 \cdot 10 = 286$$

Counting Passwords

(Reading Exercise)

- Combining the sum and product rule allows us to solve more complex problems.

Example: Each user on a computer system has a password, which is **six to eight characters** long, where each character is an **uppercase letter or a digit**. Each password must contain **at least one digit**. How many possible passwords are there?

Solution: Let P be the total number of passwords, and let P_6 , P_7 , and P_8 be the passwords of length 6, 7, and 8.

- By the sum rule $P = P_6 + P_7 + P_8$.
- To find each of P_6 , P_7 , and P_8 , we find the number of passwords of the specified length composed of letters and digits and subtract the number composed only of letters. We find that:

$$P_6 = 36^6 - 26^6 = 2,176,782,336 - 308,915,776 = 1,867,866,560.$$

$$P_7 = 36^7 - 26^7 = 78,364,164,096 - 8,031,810,176 = 70,332,353,920.$$

$$P_8 = 36^8 - 26^8 = 2,821,109,907,456 - 208,827,064,576 = 2,612,282,842,880.$$

Consequently, $P = P_6 + P_7 + P_8 = 2,684,483,063,360$.

Basic Counting Principles: Subtraction Rule

Subtraction Rule: If a task can be done either in one of n_1 ways or in one of n_2 ways, then the total number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

- Also known as, the *principle of inclusion-exclusion* (排容原理):

$$|A \cup B| = |A| + |B| - |A \cap B|$$

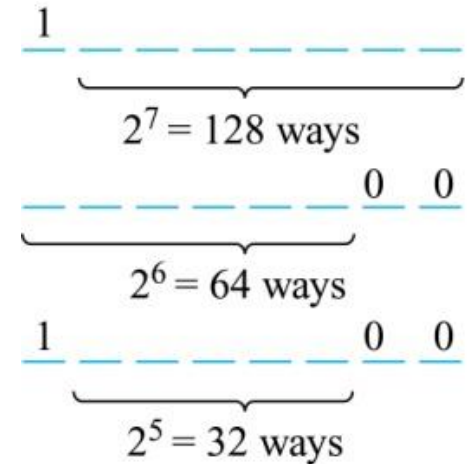
Counting Bit Strings

Example: How many bit strings of **length eight** either **start with a 1** bit **or end with** the two bits **00**?

Solution: Use the subtraction rule.

- Number of bit strings of length eight that start with a 1 bit: $2^7 = 128$
- Number of bit strings of length eight that start with bits 00: $2^6 = 64$
- Number of bit strings of length eight that start with a 1 bit and end with bits 00 : $2^5 = 32$

Hence, the number is $128 + 64 - 32 = 160$.

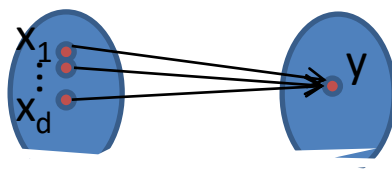


Basic Counting Principles: Division Rule

Division Rule: There are n/d ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w , exactly d of the n ways correspond to way w .

(e.g., task:數人頭; procedure:數耳朵; 每1人頭=2耳朵)

- Restated **in terms of sets**: If the finite set A (with $|A|=n$) is the union of r pairwise disjoint subsets each with d elements, then $r = n/d$.
- **In terms of functions**: If f is a function from A to B , where both are finite sets, and for every value $y \in B$ there are exactly d values $x \in A$ such that $f(x) = y$, then $|B| = |A|/d$.



Basic Counting Principles: Division Rule

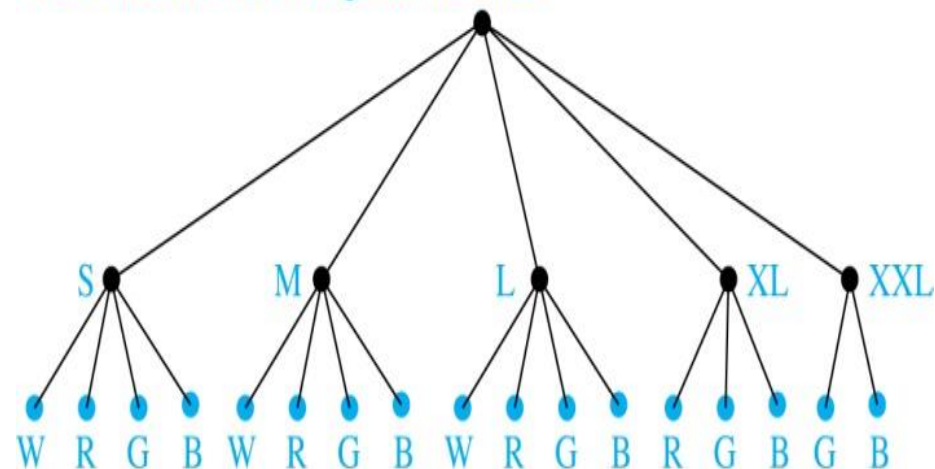
Example: How many ways are there to seat 4 people around a circular table, where two seatings are considered the same when each person has the same left and right neighbor?

Solution: Number the seats around the table from 1 to 4 proceeding clockwise. There are 4 ways to select the person for seat 1, 3 for seat 2, 2 for seat 3, and 1 way for seat 4. Thus there are $4! = 24$ ways to order the four people. But since two seatings are the same when each person has the same left and right neighbor, for every choice for seat 1, we get the 4 same seating arrangements.

Therefore, by the division rule, there are $24/4 = 6$ different seating arrangements.

Tree Diagrams

- **Tree Diagrams:** We can solve many counting problems through the use of *tree diagrams*, where a **branch** (分枝) represents a possible choice and the **leaves** (樹葉) represent possible outcomes.
- **Example:** Suppose that “I Love Discrete Math” T-shirts come in five different sizes: S,M,L,XL, and XXL. Each size comes in four colors (white, red, green, and black), except XL, which comes only in red, green, and black, and XXL, which comes only in green and black. What is the minimum number of stores that the campus book store needs to stock to have one of each size and color available?
- **Solution:** Draw the tree diagram. W = white, R = red, G = green, B = black



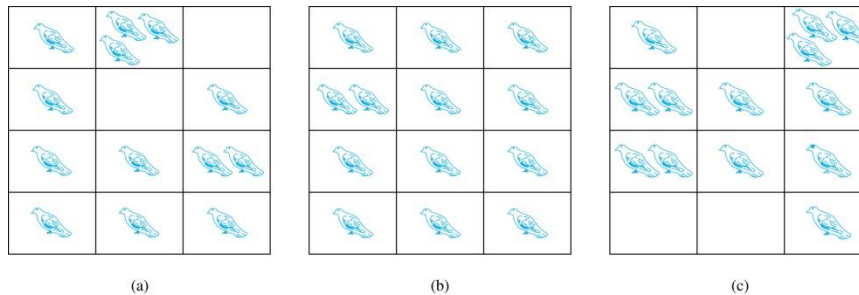
- The store must stock 17 T-shirts.

The Pigeonhole Principle

Section 6.2

The Pigeonhole Principle(鴿籠原理)

- If a flock of 13 pigeons roosts in a set of 12 pigeonholes, one of the pigeonholes must have more than 1 pigeon.



Pigeonhole Principle: If $k \in \mathbf{Z}^+$ and $k + 1$ objects are placed into k boxes, then at least one box contains two or more objects.

Proof: We use a **proof by contraposition**. Suppose none of the k boxes has more than one object. Then the total number of objects would be at most k . This contradicts the statement that we have $k + 1$ objects.



The Pigeonhole Principle

Corollary 1: A function f from a set with $k + 1$ elements to a set with k elements is not one-to-one.

Proof: Use the pigeonhole principle.

- Create a box for each element y in the codomain of f .
- Put in the box for y all of the elements x from the domain such that $f(x) = y$.
- Because there are $k + 1$ elements and only k boxes, at least one box has two or more elements.

Hence, f can't be one-to-one.



Pigeonhole Principle

Example: Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.

Example: Show that for every integer n there is a **positive multiple of n** (n 的倍數) that has **only 0s and 1s in its decimal expansion**.

Solution: Let n be a positive integer. Consider the **$n + 1$ integers**

1, 11, 111, ..., 11...1

 $n+1$ 個1

There are **n possible remainders** when an integer is **divided by n** .

By the pigeonhole principle, when each of the **$n + 1$ integers** is divided by n , **at least two must have the same remainder**.

Subtract the smaller from the larger and the result is a multiple of n that has only 0s and 1s in its decimal expansion.

The Generalized Pigeonhole Principle

The Generalized Pigeonhole Principle: If N objects are placed into k boxes, then there is at least one box containing $\geq \lceil N/k \rceil$ objects.

Proof: We use a proof by contraposition. Suppose that none of the boxes contains more than $\lceil N/k \rceil - 1$ objects. Then the total number of objects (N) is at most

$$k \left(\left\lceil \frac{N}{k} \right\rceil - 1 \right) < k \left(\left(\frac{N}{k} + 1 \right) - 1 \right) = N,$$

where the inequality $\lceil N/k \rceil < (N/k) + 1$ has been used.
That is $N < N$ (**contradiction!**) 

Example: Among 100 people there are at least $\lceil 100/12 \rceil = 9$ who were born in the same month.

The Generalized Pigeonhole Principle

Example: a) How many cards must be selected from a standard deck of 52 cards to guarantee that at least 3 cards of the same suit (花色) are chosen?

b) How many must be selected to guarantee that at least 3 hearts (紅心) are selected?

Solution:

a) We assume four boxes; one for each suit. Using the generalized pigeonhole principle, at least one box contains at least $\lceil N/4 \rceil$ cards. At least three cards of one suit are selected if $\lceil N/4 \rceil \geq 3$. The smallest integer N such that $\lceil N/4 \rceil \geq 3$ is $N = 2 \cdot 4 + 1 = 9$.

b) A deck contains 13 hearts and 39 cards which are not hearts. So, if we select 41 cards, we may have 39 cards which are not hearts along with 2 hearts. However, when we select 42 cards, we must have at least three hearts. (Note that the generalized pigeonhole principle is not used here.)

Permutations and Combinations (排列組合)

Section 6.3

Permutations

Definition: A *permutation* of a set of distinct objects is an ordered arrangement of these objects.

An ordered *arrangement of r elements* of a set is called an *r -permutation*.

Example: Let $S = \{1,2,3\}$.

- The ordered arrangement 3,1,2 is a permutation of S .
- The ordered arrangement 3,2 is a 2-permutation of S .
- The number of r -permutations of a set with n elements is denoted by $P(n,r)$.
 - The 2-permutations of $S = \{1,2,3\}$ are 1,2; 1,3; 2,1; 2,3; 3,1; and 3,2. Hence, $P(3,2) = 6$.

A Formula for the Number of Permutations

Theorem 1: If n is a positive integer and r is an integer with $1 \leq r \leq n$, then there are

$$P(n, r) = n(n - 1)(n - 2) \cdots (n - r + 1)$$

r -permutations of a set with n distinct elements.

Proof: Use the **product rule**. The first element can be chosen in n ways. The second in $n - 1$ ways, and so on until there are $(n - (r - 1))$ ways to choose the last element.

- Note that $P(n, 0) = 1$, since there is only one way to order zero elements.

Corollary 1: If n and r are integers with $1 \leq r \leq n$, then

$$P(n, r) = \frac{n!}{(n-r)!}$$

Solving Counting Problems by Counting Permutations

Example: How many ways are there to select a **first-prize** winner, a **second prize** winner, and a **third-prize** winner from 100 different people who have entered a contest?

Solution:

$$P(100,3) = 100 \cdot 99 \cdot 98 = 970,200$$

Solving Counting Problems by Counting Permutations (*continued*)

Example: How many permutations of the letters *ABCDEFGH* contain the string *ABC* ?

Solution: We solve this problem by counting the permutations of six objects, *ABC*, *D*, *E*, *F*, *G*, and *H*.

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

Combinations

Definition: An *r -combination* of elements of a set is an *unordered* selection of r elements from the set. Thus, an r -combination is simply a subset of the set with r elements.

- The number of r -combinations of a set with n distinct elements is denoted by $C(n, r)$. The notation $\binom{n}{r}$ is also used and is called a *binomial coefficient*. (We will see the notation again in the binomial theorem in Section 6.4.)

Example: Let S be the set $\{a, b, c, d\}$. Then $\{a, c, d\}$ is a 3-combination from S . It is the same as $\{d, c, a\}$ since the order listed does not matter.

- $C(4, 2) = 6$ because the 2-combinations of $\{a, b, c, d\}$ are the six subsets $\{a, b\}$, $\{a, c\}$, $\{a, d\}$, $\{b, c\}$, $\{b, d\}$, and $\{c, d\}$.

Combinations

Theorem 2: The number of r -combinations of a set with n elements, where $n \geq r \geq 0$, equals

$$C(n, r) = \frac{n!}{(n-r)!r!}.$$

Proof: By the product rule $P(n, r) = C(n, r) \cdot P(r, r)$. Therefore,

$$C(n, r) = \frac{P(n, r)}{P(r, r)} = \frac{n!/(n-r)!}{r!/(r-r)!} = \frac{n!}{(n-r)!r!}.$$

Combinations

Example: How many poker hands of five cards can be dealt from a standard deck of 52 cards? Also, how many ways are there to select 47 cards from a deck of 52 cards?

Solution: Since the order in which the cards are dealt does not matter, the number of five card hands is:

$$\begin{aligned} C(52, 5) &= \frac{52!}{5!47!} \\ &= \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 26 \cdot 17 \cdot 10 \cdot 49 \cdot 12 = 2,598,960 \end{aligned}$$

- The different ways to select 47 cards from 52 is

$$C(52, 47) = \frac{52!}{47!5!} = C(52, 5) = 2,598,960.$$

This is a special case of a general result. →

Combinations

Corollary 2: Let n and r be nonnegative integers with $r \leq n$. Then $C(n, r) = C(n, n - r)$.

Proof: From Theorem 2, it follows that

$$C(n, r) = \frac{n!}{(n-r)!r!}$$

and

$$C(n, n - r) = \frac{n!}{(n-r)![n-(n-r)]!} = \frac{n!}{(n-r)!r!} .$$



Hence, $C(n, r) = C(n, n - r)$.

This result can be proved without using algebraic manipulation. →

Combinatorial Proofs

Definition 1: A *combinatorial proof* of an identity is a proof that uses one of the following methods.

- A *double counting proof* uses counting arguments to prove that both sides of an identity **count the same objects, but in different ways**. (等式兩邊要算的東西是一樣的只是方法不同)
- A *bijection proof* shows that there is a **bijection between the sets of objects counted by the two sides of the identity**. (等式兩邊要算的東西雖不一樣，但兩者間存在一對一且映成的函數)

*註: combinatorial proof (組合證明)，也就是靠計算數量而得的證明。

Combinatorial Proofs

- Here are two combinatorial proofs that

$$C(n, r) = C(n, n - r)$$

when r and n are nonnegative integers with $r < n$:

Proof:

Suppose that S is a set with n elements.

By definition the number of subsets of S with r elements is $C(n, r)$.

– *Bijjective Proof:*

The function that maps a subset A of S to $\bar{A}$ is a bijection between the subsets of S with r elements and the subsets with $n - r$ elements.

By definition the number of subsets of S with $n - r$ elements is $C(n, n - r)$. Since there is a bijection between the two sets, they must have the same number of elements. ◀

- *Double Counting Proof:* Each subset A of S can also be described by specifying which elements are not in A , i.e., those which are in $\bar{A}$. Since the complement of a subset of S with r elements has $n - r$ elements, there are also $C(n, n - r)$ subsets of S with r elements. ◀

Combinations

Example: How many ways are there to select 5 players from a 10-member tennis team to make a trip to a match at another school.

Solution: By Theorem 2, the number of combinations is

$$C(10, 5) = \frac{10!}{5!5!} = 252.$$

Example: A group of 30 people have been trained as astronauts to go on the first mission to Mars. How many ways are there to select a crew of six people to go on this mission?

Solution: By Theorem 2, the number of possible crews is

$$C(30, 6) = \frac{30!}{6!24!} = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26 \cdot 25}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 593,775 .$$

Binomial Coefficients and Identities

Section 6.4

Powers of Binomial Expressions

Definition: A *binomial expression* is the sum of two terms, such as $x + y$.

- We can use counting principles to find the **coefficients** (係數) in the expansion (展開式) of $(x + y)^n$ where n is a positive integer.
- To illustrate this idea, we first look at the process of expanding $(x + y)^3$.
- $(x + y)(x + y)(x + y)$ expands into a sum of terms that are the product of a term from each of the three sums. (和之積=>積之和)
- Terms of the form x^3, x^2y, xy^2, y^3 arise. The question is what are the coefficients?
 - To obtain x^3 , an x must be chosen from each of the sums. There is only one way to do this. So, the coefficient of x^3 is 1.
 - To obtain x^2y , an x must be chosen from two of the sums and a y from the other. There are $\binom{3}{2}$ ways to do this and so the coefficient of x^2y is 3.
 - To obtain xy^2 , an x must be chosen from one of the sums and a y from the other two. There are $\binom{3}{1}$ ways to do this and so the coefficient of xy^2 is 3.
 - To obtain y^3 , a y must be chosen from each of the sums. There is only one way to do this. So, the coefficient of y^3 is 1.
- We have used a counting argument to show that $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$.
- Next we present the binomial theorem gives the coefficients of the terms in the expansion of $(x + y)^n$.

Binomial Theorem

Binomial Theorem: Let x and y be variables, and n a nonnegative integer. Then:

$$(x+y)^n = \sum_{j=0}^n \binom{n}{j} x^{n-j} y^j = \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \cdots + \binom{n}{n-1} x y^{n-1} + \binom{n}{n} y^n.$$

Proof: We use **combinatorial reasoning**.

The terms in the expansion of $(x + y)^n$ are of the form $x^{n-j}y^j$ for $j = 0, 1, 2, \dots, n$.

To form the term $x^{n-j}y^j$, it is necessary to choose $n-j$ x s from the n sums. Therefore, the coefficient of $x^{n-j}y^j$ is $\binom{n}{n-j}$ which equals $\binom{n}{j}$. ◀

Using the Binomial Theorem

Example: What is the coefficient of $x^{12}y^{13}$ in the expansion of $(2x - 3y)^{25}$?

Solution: We view the expression as $(2x + (-3y))^{25}$.

By the binomial theorem

$$(2x + (-3y))^{25} = \sum_{j=0}^{25} \binom{25}{j} (2x)^{25-j} (-3y)^j.$$

Consequently, the coefficient of $x^{12}y^{13}$ in the expansion is obtained when $j = 13$. That is

$$\binom{25}{13} 2^{12} (-3)^{13} = -\frac{25!}{13!12!} 2^{12} 3^{13}.$$

A Useful Identity

Corollary 1: With $n \geq 0$, $\sum_{k=0}^n \binom{n}{k} = 2^n$.

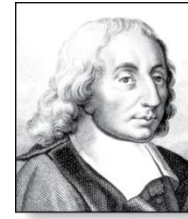
Proof (using binomial theorem): With $x = 1$ and $y = 1$, from the binomial theorem we see that:

$$2^n = (1 + 1)^n = \sum_{k=0}^n \binom{n}{k} 1^k 1^{(n-k)} = \sum_{k=0}^n \binom{n}{k}.$$

Proof (combinatorial): Consider the subsets of a set with n elements. There are $\binom{n}{0}$ subsets with zero elements, $\binom{n}{1}$ with one element, $\binom{n}{2}$ with two elements, ..., and $\binom{n}{n}$ with n elements. Therefore the total is $\sum_{k=0}^n \binom{n}{k}$.

Since, we know that a set with n elements has 2^n subsets, we conclude: $\sum_{k=0}^n \binom{n}{k} = 2^n$.

Pascal's Identity



Blaise Pascal
(1623-1662)

Pascal's Identity: If n and k are integers with $n \geq k \geq 0$, then

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}.$$

Proof (combinatorial): Let T be a set where $|T| = n + 1$, $a \in T$, and $S = T - \{a\}$. There are $\binom{n+1}{k}$ subsets of T containing k elements. Each of these subsets **either**:

- **contains a** together with $k - 1$ other elements of S , or
- contains k elements of S and **not contain a** .

There are

- $\binom{n}{k-1}$ subsets of k elements that **contain a** , since there are $\binom{n}{k-1}$ subsets of $k - 1$ elements of S ,
- $\binom{n}{k}$ subsets of k elements of T that **do not contain a** , because there are $\binom{n}{k}$ subsets of k elements of S .

Hence, $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$. ◀

Pascal's Triangle

The n th row in the triangle consists of the binomial coefficients $\binom{n}{k}$, $k = 0, 1, \dots, n$.

$$\begin{array}{c}
 \binom{0}{0} \\
 \binom{1}{0} \quad \binom{1}{1} \\
 \binom{2}{0} \quad \binom{2}{1} \quad \binom{2}{2} \\
 \binom{3}{0} \quad \binom{3}{1} \quad \binom{3}{2} \quad \binom{3}{3} \\
 \binom{4}{0} \quad \binom{4}{1} \quad \binom{4}{2} \quad \binom{4}{3} \quad \binom{4}{4} \\
 \binom{5}{0} \quad \binom{5}{1} \quad \binom{5}{2} \quad \binom{5}{3} \quad \binom{5}{4} \quad \binom{5}{5} \\
 \binom{6}{0} \quad \binom{6}{1} \quad \binom{6}{2} \quad \binom{6}{3} \quad \binom{6}{4} \quad \binom{6}{5} \quad \binom{6}{6} \\
 \binom{7}{0} \quad \binom{7}{1} \quad \binom{7}{2} \quad \binom{7}{3} \quad \binom{7}{4} \quad \binom{7}{5} \quad \binom{7}{6} \quad \binom{7}{7} \\
 \binom{8}{0} \quad \binom{8}{1} \quad \binom{8}{2} \quad \binom{8}{3} \quad \binom{8}{4} \quad \binom{8}{5} \quad \binom{8}{6} \quad \binom{8}{7} \quad \binom{8}{8} \\
 \dots \\
 \text{(a)}
 \end{array}$$

By Pascal's identity:

$$\binom{6}{4} + \binom{6}{5} = \binom{7}{5}$$

$$\begin{array}{c}
 1 \\
 1 \quad 1 \\
 1 \quad 2 \quad 1 \\
 1 \quad 3 \quad 3 \quad 1 \\
 1 \quad 4 \quad 6 \quad 4 \quad 1 \\
 1 \quad 5 \quad 10 \quad 10 \quad 5 \quad 1 \\
 1 \quad 6 \quad 15 \quad 20 \quad 15 \quad 6 \quad 1 \\
 1 \quad 7 \quad 21 \quad 35 \quad 35 \quad 21 \quad 7 \quad 1 \\
 1 \quad 8 \quad 28 \quad 56 \quad 70 \quad 56 \quad 28 \quad 8 \quad 1 \\
 \dots \\
 \text{(b)}
 \end{array}$$

By Pascal's identity, adding two adjacent binomial coefficients results in the binomial coefficient in the next row between these two coefficients.

Generalized Permutations and Combinations

Section 6.5

Permutations with Repetition

Theorem 1: The number of r -permutations of a set of n objects with **repetition allowed** is n^r .

Proof: There are n ways to select an element of the set for each of the r positions in the r -permutation when repetition is allowed. Hence, by the **product rule** there are n^r r -permutations with repetition.



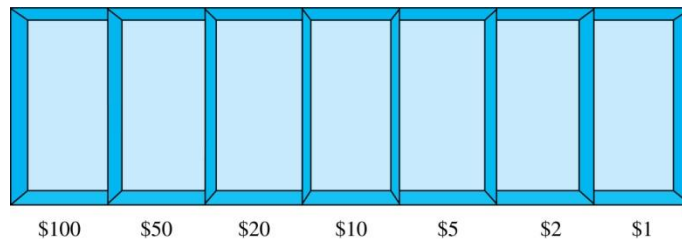
Example: How many **strings** of **length r** can be formed from the uppercase letters of the English alphabet?

Solution: The number of such strings is 26^r , which is the number of r -permutations of a set with 26 elements.

Combinations with Repetition

Example: How many ways are there to **select five bills** from a box containing at least five of each of the following denominations: \$1, \$2, \$5, \$10, \$20, \$50, and \$100?

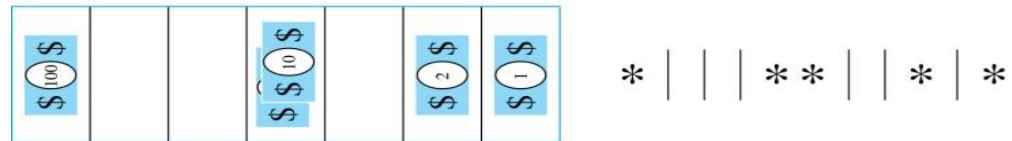
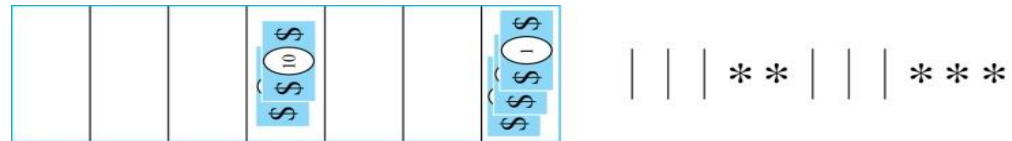
Solution: Place the selected bills in the appropriate position of a cash box illustrated below:



continued →

Combinations with Repetition

- Some possible ways of placing the five bills:



- The number of ways to select five bills corresponds to the number of ways to arrange six bars and five stars in a row.
- This is the number of unordered selections of 5 objects from a set of 11. Hence, there are

$$C(11, 5) = \frac{11!}{5!6!} = 462$$

ways to choose five bills with seven types of bills.

Combinations with Repetition

Theorem 2: The number of r -combinations from a set with n elements when repetition of elements is allowed is

$$C(n + r - 1, r) = C(n + r - 1, n - 1).$$

Proof: Each r -combination of a set with n elements with repetition allowed can be represented by a list of $n - 1$ bars and r stars. The bars mark the n cells containing a star for each time the i th element of the set occurs in the combination.

The number of such lists is $C(n + r - 1, r)$, because each list is a choice of the r positions to place the stars, from the total of $n + r - 1$ positions to place the stars and the bars. This is also equal to $C(n + r - 1, n - 1)$, which is the number of ways to place the $n - 1$ bars.



Combinations with Repetition

Example: How many solutions does the equation

$$x_1 + x_2 + x_3 = 11$$

have, where x_1 , x_2 and x_3 are **nonnegative integers**?

Solution: Each solution corresponds to **a way to select 11 items from a set with three elements**; x_1 elements of type one, x_2 of type two, and x_3 of type three.

By Theorem 2 it follows that there are

$$C(3 + 11 - 1, 11) = C(13, 11) = C(13, 2) = \frac{13 \cdot 12}{1 \cdot 2} = 78$$

solutions.



Combinations with Repetition

Example: Suppose that a cookie shop has 4 different kinds of cookies. How many different ways can 6 cookies be chosen?

Solution: The number of ways to choose six cookies is the number of 6-combinations of a set with four elements. By Theorem 2

$$C(9, 6) = C(9, 3) = \frac{9 \cdot 8 \cdot 7}{1 \cdot 2 \cdot 3} = 84$$

is the number of ways to choose six cookies from the four kinds.

Summarizing the Formulas for Counting Permutations and Combinations with and without Repetition

TABLE 1 Combinations and Permutations With and Without Repetition.

<i>Type</i>	<i>Repetition Allowed?</i>	<i>Formula</i>
r -permutations	No	$\frac{n!}{(n - r)!}$
r -combinations	No	$\frac{n!}{r! (n - r)!}$
r -permutations	Yes	n^r
r -combinations	Yes	$\frac{(n + r - 1)!}{r! (n - 1)!}$

Permutations with Indistinguishable Objects

Example: How many different strings can be made by reordering the letters of the word *SUCCESS*.

Solution: There are seven possible positions for the three Ss, two Cs, one U, and one E.

- The three Ss can be placed in $C(7,3)$ different ways, leaving four positions free.
- The two Cs can be placed in $C(4,2)$ different ways, leaving two positions free.
- The U can be placed in $C(2,1)$ different ways, leaving one position free.
- The E can be placed in $C(1,1)$ way.

By the product rule, the number of different strings is:

$$C(7,3)C(4,2)C(2,1)C(1,1) = \frac{7!}{3!4!} \cdot \frac{4!}{2!2!} \cdot \frac{2!}{1!1!} \cdot \frac{1!}{1!0!} = \frac{7!}{3!2!1!1!} = 420.$$

The reasoning can be generalized to the following theorem. →

Permutations with Indistinguishable Objects

Theorem 3: The number of different permutations of n objects, where there are n_1 indistinguishable objects of type 1, n_2 indistinguishable objects of type 2, ..., and n_k indistinguishable objects of type k , is:

$$\frac{n!}{n_1!n_2!\cdots n_k!} \cdot$$

Proof: By the product rule the total number of permutations is:

$C(n, n_1) C(n - n_1, n_2) \cdots C(n - n_1 - n_2 - \cdots - n_k, n_k)$ since:

- The n_1 objects of type one can be placed in the n positions in $C(n, n_1)$ ways, leaving $n - n_1$ positions.
- Then the n_2 objects of type two can be placed in the $n - n_1$ positions in $C(n - n_1, n_2)$ ways, leaving $n - n_1 - n_2$ positions.
- Continue in this fashion, until n_k objects of type k are placed in $C(n - n_1 - n_2 - \cdots - n_k, n_k)$ ways.

The product can be manipulated into the desired result as follows:

$$\frac{n!}{n_1!(n-n_1)!} \frac{(n-n_1)!}{n_2!(n-n_1-n_2)!} \cdots \frac{(n-n_1-\cdots-n_{k-1})!}{n_k!0!} = \frac{n!}{n_1!n_2!\cdots n_k!} \cdot$$



Distributing Objects into Boxes

- Many counting problems can be solved by counting the ways objects can be placed into boxes.

- We can consider two types of objects:

1. Distinguishable objects (e.g., “Julie, Vivian, and Wilson”)



2. Indistinguishable objects (e.g., “three students”)



- We can also consider two types of “boxes”

1. Distinguishable boxes (e.g., “room G313 and room G508”)



2. Indistinguishable boxes (e.g., “two homerooms”)



Distributing Objects into Boxes

- *Distinguishable objects / distinguishable boxes*

- Example: How many ways are there to deal 5-card poker hands from a 52-card deck to each of four players?

Solutions:

- Player 1: $C(52,5)$ ways to deal
- Player 2: $C(47,5)$ ways to deal
- Player 3: $C(42,5)$ ways to deal
- Player 4: $C(37,5)$ ways to deal



Based on the product rule, there are
 $C(52,5) \times C(47,5) \times C(42,5) \times C(37,5)$
 $= 52! / (5!5!5!5!32!)$
ways to distribute hands of 5 cards to each of 4 players.

Theorem 4: The number of ways to distribute n distinguishable objects into k distinguishable boxes so that n_i objects are placed into box i , $i=1,2,\dots,k$, equals

$$\frac{n!}{n_1!n_2!\cdots n_k!} \cdot$$

- (See Exercises 47 and 48 for two different proofs.)

Distributing Objects into Boxes

- *Indistinguishable objects / distinguishable boxes*

- This turns out to be the same as counting the n -combinations for a set with k elements when repetition is allowed!(ppt.p50)

Recall: We solved the above problem by arranging placeholders (*) and dividers (|).
To place n indistinguishable items into k distinguishable bins:

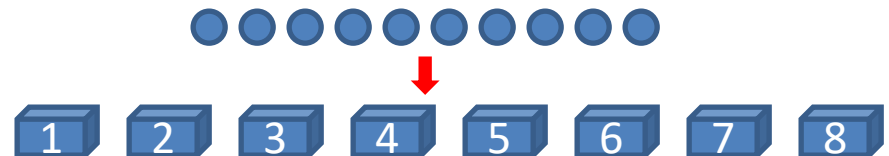
1. Treat our indistinguishable items as *s
2. Use | to divide our distinguishable bins
3. Count the ways to arrange n placeholders and $k-1$ dividers

Result: There are $C(n + k - 1, n)$ ways to place n indistinguishable objects into k distinguishable boxes

- **Example:** How many ways are there to place 10 indistinguishable balls into 8 distinguishable bins?

Observation:

1. Treat balls as *s
2. Use $8-1 = 7$ dividers to separate bins
3. Pick 10 positions out of a total 17 to place balls (all remaining positions will be bin dividers)



Solution: We have $C(8 + 10 - 1, 10) = C(17, 10) = 19,448$ ways to place 10 indistinguishable objects into 8 distinguishable boxes.

Distributing Objects into Boxes

- *Distinguishable objects and indistinguishable boxes.*

- Example: How many ways are there to put 4 different employees into 3 indistinguishable offices, when each office can contain any number of employees?

Solution:

Let's call the 4 employees A, B, C, and D

Goal: Partition A, B, C, and D into at most 3 disjoint subsets

One way to put everyone in the same office

$\{\{A, B, C, D\}\}$

Seven ways to put everyone in 2 offices

$\{\{A, B, C\}, \{D\}\}$, $\{\{A, B, D\}, \{C\}\}$, $\{\{A, C, D\}, \{B\}\}$, $\{\{B, C, D\}, \{A\}\}$
 $\{\{A, B\}, \{C, D\}\}$, $\{\{A, C\}, \{B, D\}\}$, $\{\{A, D\}, \{B, C\}\}$

Six ways to put everyone into 3 offices

$\{\{A, B\}, \{C\}, \{D\}\}$, $\{\{A, C\}, \{B\}, \{D\}\}$, $\{\{A, D\}, \{B\}, \{C\}\}$
 $\{\{B, C\}, \{A\}, \{D\}\}$, $\{\{B, D\}, \{A\}, \{C\}\}$, $\{\{C, D\}, \{A\}, \{B\}\}$

Total: 14 ways to assign 4 different employees to 3 indistinguishable offices

- Is there a **simple closed formula** for the number of ways to distribute n distinguishable objects into k indistinguishable boxes?

No! but there is a complicated one

$S(n, j)$ is a **Stirling number of the second kind** that tells us the number of ways that a set of n items can be partitioned into j **non-empty** subsets.

$S(n, j)$ is defined as follows:

$$S(n, j) = \frac{1}{j!} \sum_{i=0}^{j-1} (-1)^i C(j, i) (j-i)^n$$

Result: The number of ways to distribute n distinguishable objects into k indistinguishable boxes is:

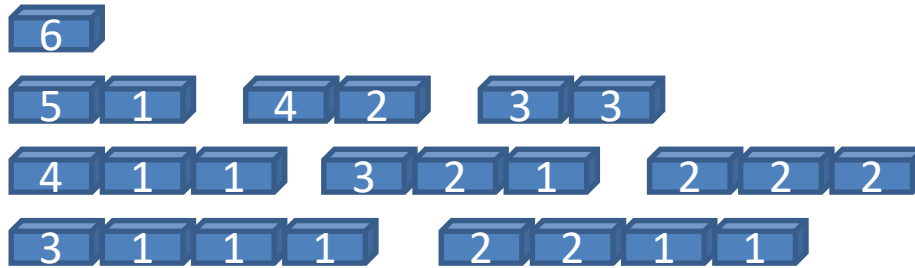
$$\sum_{j=1}^k S(n, j) = \sum_{j=1}^k \frac{1}{j!} \sum_{i=0}^{j-1} (-1)^i C(j, i) (j-i)^n$$

Distributing Objects into Boxes

- *Indistinguishable objects / indistinguishable boxes.*

- Example: How many ways can six copies of the same book be packed in at most four boxes, if each box can hold up to six books?

Solution:



Total: There are 9 ways to pack 6 identical books into at most 4 indistinguishable boxes.

- better way to do this? Unfortunately no!

Here's why: Placing n indistinguishable objects into k indistinguishable boxes is **the same as** writing n as the sum of at most k positive integers arranged in non-increasing order.

i.e., $n = a_1 + a_2 + \dots + a_j$, where $a_1 \geq a_2 \geq \dots \geq a_j$, and $j \leq k$

We say that $a_1, a_2, \dots, a_j$ is **a partition of n into j integers**

There is **no simple closed formula** for counting the partitions of an integer, thus there is no solution for placing n indistinguishable items into k indistinguishable boxes.

Generating Permutations and Combinations

- Sometimes permutations and combinations need to **be generated, not just being counted!**
- Ex1: [salesman problems]
visiting 6 cities without duplication with minimal instances.
==> try all possible permutations of the cities and find one with minimal instances.
- Ex2: $S = \{x_1, \dots, x_6\}$, find a subset $\{y_1, \dots, y_j\}$ of S s.t. $\sum y_k = 100$.
Sol: find each subset (combination) T of S and check their sum.

Generating Permutations and Combinations

- **Generating permutations** based on lexicographic ordering of the set of permutation of $\{1,2,3,\dots,n\}$
 - $a_1a_2\dots a_n$ **precedes** $b_1b_2\dots b_n$ if $\exists k, 1 \leq k \leq n$ s.t. $a_1=b_1, a_2=b_2, \dots, a_{k-1}=b_{k-1}$, and $a_k < b_k$.
 - E.g., **234**15 precedes **235**14;
 - 4**1523 precedes **5**2143
 - Sketch of the algorithm to generate the **next** permutation:
 - Given a permutation
 - Choose the right-most pair a_j, a_{j+1} where $a_j < a_{j+1}$
 - Choose the **least items** to the right of a_j **greater than a_j**
 - Swap this item and a_j
 - **Arrange the remaining (to the right) items in order**
 - E.g., Generate the permutation of the integers 1, 2, 3 in lexicographic order.

3 6 2 5 4 1
3 6 2 < 5 > 4 > 1
3 6 2 < 5 > <u>4</u> > 1
3 6 4 < 5 > <u>2</u> > 1
3 6 4 <u>1 2 5</u>

Sol: 1 **2** < 3 → 1 < 3 2 → 2 1 < 3 → 2 < 3 1 → 3 1 < 2 → 3 2 1

Generating Permutations and Combinations

- Generating communications

- Generate **all the combinations** of the elements of a finite set:

$\{a_1, a_2, \dots, a_k, \dots, a_{n-1}, a_n\}$								
0	0	...	0	...	0	0		ϕ
0	0	...	0	...	0	1		$\{a_n\}$
0	0	...	0	...	1	0		$\{a_{n-1}\}$
...			...			...		...
1	1	...	1	...	1	1		$\{a_1, a_2, \dots, a_k, \dots, a_{n-1}, a_n\}$

- Generate the r-combination of the set $\{1, 2, 3, \dots, n\}$

4-combination of $\{1, 2, 3, 4, 5, 6\}$
$\{1, 2, 3, 4\}$
...
$\{1, \mathbf{2}, 5, 6\}$
$\{1, \mathbf{3}, 5, 6\}$
$\{1, \mathbf{3}, \mathbf{4}, 5\}$

 - Start with $\{1, \dots, r\}$
 - To find the next combination of $a_1 a_2 \dots a_r$
 - Locate the last element a_i such that $a_i \neq n-r+i$
 - Replace a_i with a_i+1
 - Replace a_j with a_i+j-i for $j=i+1, i+2, \dots, r$